



DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY
UNIVERSITY SUPPLEMENTARY/SPECIAL EXAMINATION FOR 2024/2025
ACADEMIC YEAR

BSc. MECHATRONICS ENGINEERING and BSc. GEGIS.

SMA 3244 : PARTIAL DIFFERENTIAL EQUATIONS

INSTRUCTIONS: ANSWER QUESTION ONE AND ANY OTHER TWO QUESTIONS

QUESTION ONE-30 MARKS-COMPUSORY

6/8/2023

11.00-01.00PM

- a) Define a P.D.E. and state two methods of its formation. (2 Mark)
- b) Formulate the PDE from the function $F(x^3 - y^3, x^2 - z^2) = 0$ (4 Marks)
- c) Find the equation of the tangent to the sphere whose equation is given by $f(x, y, z) = x^2 + y^2 + z^2 - 9 = 0$ at the point $p(2, 1, 2)$. (5 Marks)
- d) Find the integral surfaces of the equation $\frac{dx}{x} = \frac{dy}{y+z} = \frac{dz}{x^2+z}$ (5 Marks)
- e) Use a suitable substitution to solve the homogeneous differential equation $(y^2 + yz)dx + (xz + z^2)dy + (y^2 - xy)dz = 0$. (5 Marks)
- f) Solve the equation defined by $y^2 zp - x^2 zq - x^2 y = 0$. (5 Marks)
- g) Determine the Laplace transform of $\frac{\partial U}{\partial t} + \frac{\partial^2 U}{\partial x^2} = 0$ (4 marks)

QUESTION TWO: 20 MARKS:

- a) Obtain the partial differential equation whose solutions is given by $F(u, v) = 0$, where u and v are functions of x, y and z , while z is a function of x and y . (9 Marks)
- b) Solve the given equations;
- i) $\frac{dx}{xy} = \frac{dy}{y^2} = \frac{dz}{zxy - 2x^2}$ (5 Marks)
- ii) $\frac{dx}{x^2(y^3 - z^3)} = \frac{dy}{y^2(z^3 - x^3)} = \frac{dz}{z^2(x^3 - y^3)}$ (6 Marks)

QUESTION THREE: 20 MARKS:

- a) Solve the heat equation, $\frac{\partial u}{\partial t} - \alpha \frac{\partial^2 u}{\partial x^2} = 0$ by the method of separation of variables, for
 $u(0,t) = u(4,t) = 0$ and $u(x,0) = x - 4$. (13 Marks)
- b) Show that $ydx + xdy + 2zdz = 0$, is integrable hence solve by a suitable substitution. (7 Marks)

QUESTION FOUR: 20 MARKS:

- a) Find the Orthogonal Trajectories of the surface $x^2 + y^2 + z^2 = a^2$, with its intersection with the family of surface $xy = cz$. (10 Marks)
- b) Show that the simultaneous equations: $p_1 dx + Q_1 dy + R_1 dz = 0$
 $p_2 dx + Q_2 dy + R_2 dz = 0$ Can be reduced to take the form of: $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$ Hence solve: $\frac{dx}{x+z} = \frac{dy}{y} = \frac{dz}{z+y^2}$ (10 Marks)

QUESTION FIVE: 20 MARKS:

- a) Solve $\frac{\partial^2 u}{\partial t^2} = 4 \frac{\partial^2 u}{\partial x^2}$, using Laplace transforms given the boundary and initial conditions
 $u(0,t) = 0$ $u(x,0) = 0$
 $u(l,t) = 0$ $u_t(x,0) = \sin \pi x$ (12 Marks)
- b) Find the integral curves of the equation $\frac{dx}{y(x+y)+az} = \frac{dy}{x(x+y)-az} = \frac{dz}{(x+y)z}$ (8 marks)